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Brain age gap as a predictive biomarker that links aging, lifestyle, and neuropsychiatric health

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Abstract

Background The brain age gap (BAG) is a neuroimaging-derived marker of accelerated brain aging. However, its clinical application faces challenges due to model inaccuracies and unclear links to disease mechanisms. This study investigates the clinical relevance of BAG across neuropsychiatric disorders, cognitive decline, mortality, and lifestyle interventions.

Methods We use data from multiple cohorts, including 38,967 participants from the UK Biobank (ages 45–82, 52.5% female), 1,402 individuals from the ADNI study (ages 55–96, 56.0% female), and 1,182 from the PPMI study (ages 45–83, 58.0% female). We develop a 3D Vision Transformer for whole-brain age estimation. Survival analysis, restricted cubic splines, and regression models assess BAG's associations with cognitive, neuropsychiatric disorders, mortality and impact of lifestyle factors.

Results Here we show that the model achieves a mean error of 2.68 years in the UK Biobank and 2.99–3.20 years in ADNI/PPMI. Each one-year increase in BAG raises Alzheimer's risk by 16.5%, mild cognitive impairment by 4.0%, and all-cause mortality by 12%. The highest-risk group (Q4) shows a 2.8-fold increased risk of Alzheimer's disease, a 6.4-fold risk of multiple sclerosis, and a 2.4-fold higher mortality risk. Cognitive decline is most evident in Q4, particularly in reaction time and processing speed. Lifestyle interventions, especially smoking cessation, moderate alcohol consumption, and physical activity, significantly slow BAG progression in individuals with advanced neurodegeneration.

Conclusions BAG predicts accelerated brain aging, neuropsychiatric disorders, and mortality. Its ability to detect nonlinear cognitive thresholds and modifiability through lifestyle changes makes it useful for risk stratification and prevention.

Plain language summary

This study examines whether the brain age gap (BAG)—the difference between a person's estimated brain age and their chronological age—can predict risks of cognitive decline, mental health disorders, and early death. We analyzed MRI scans from over 40,000 participants across three large-scale cohort studies and used a deep-learning model to estimate brain age. We found that a larger BAG was linked to higher risks of cognitive decline, dementia, multiple sclerosis, and reduced survival. Importantly, lifestyle changes such as quitting smoking, moderate alcohol use, and regular exercise significantly reduce brain aging, especially in high-risk individuals. Measuring BAG could enable early detection of at-risk individuals and guide targeted lifestyle interventions and public health strategies to preserve brain health.

The accelerated aging of global populations has precipitated a marked increase in the incidence of age-associated neurodegenerative disorders[1–5] (e.g., Alzheimer's disease) and psychiatric conditions (e.g., depression, anxiety disorders), presenting a critical public health challenge demanding urgent resolution. Addressing this challenge necessitates elucidation of the interplay between cerebral aging processes and neurodegenerative disease mechanisms. Conventional chronological metrics inadequately capture the heterogeneity of individual brain aging, whereas the neuroimaging-derived brain age construct, quantified via machine learning algorithms, provides precise measurement of biological brain aging. The divergence between predicted brain age and chronological age, termed the Brain Age Gap (BAG)[6–8], has emerged as a promising biomarker for assessing cerebral health status, with robust empirical evidence demonstrating its association with cognitive decline[9,10], neuropathological alterations[11–14], and all-cause mortality risk.

Despite its considerable clinical potential as a quantitative index of brain aging, the translational application of BAG faces several challenges. Limitations in current predictive models[15–17]—including insufficient accuracy, unclear disease-specific aging mechanisms, and a paucity of systematic investigations into BAG trajectories—hinder its utility for early risk

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prediction and precision interventions. Notably, brain aging is modulated by multidimensional factors encompassing genetic predisposition[18,19], environmental influences[20], and lifestyle behaviors[21,22]. Empirical evidence[23–25] indicates that adverse factors such as sleep deprivation, sedentary behavior, and high-fat diets may exacerbate brain aging by promoting neuroinflammation and oxidative stress, whereas beneficial lifestyle practices (e.g., regular exercise, cognitive training, and Mediterranean diet) may confer neuroprotection by enhancing synaptic plasticity and neurogenesis. These regulatory mechanisms offer critical insights for developing targeted interventions. Although cross-sectional studies[26–28] consistently report elevated BAG in neuropsychiatric disorders—suggesting its role as a transdiagnostic marker of shared pathology—the specificity and sensitivity of BAG in predicting individual disease risks remain poorly characterized. Crucially, systematic evidence is lacking regarding BAG's capacity to serve as a biomarker for disease progression and therapeutic response.

While the association between brain age and neuropsychiatric risk garners substantial attention, the predictive utility of BAG in age-related disorders remains incompletely understood. Key unresolved questions include whether BAG can reliably predict neuropsychiatric risk and effectively monitor disease progression or treatment efficacy. Furthermore, the relationship between BAG and multidimensional cognitive functions, as well as the mechanistic basis of lifestyle interventions in attenuating brain aging, warrants further investigation. To address these knowledge gaps, this study leverages a large-scale population cohort to systematically evaluate the association between BAG and critical clinical outcomes, including cognitive decline, neuropsychiatric risk, and all-cause mortality, while examining the moderating effects of healthy lifestyle factors on brain aging trajectories. Methodologically, we employ an innovative 3D Vision Transformer (3D-ViT)[6,29,30] deep learning framework to enhance the precision and robustness of brain age estimation, enabling the identification of biologically meaningful neuroimaging markers of aging. In this study, we systematically evaluate the clinical relevance of BAG using a deep-learning model across large-scale multicohort datasets. Our findings demonstrate that an increased BAG strongly predicts higher risks of cognitive decline, neuropsychiatric disorders, and mortality. Importantly, we find that healthy lifestyle interventions effectively mitigate brain aging trajectories, highlighting BAG's potential clinical utility for personalized risk assessment and targeted preventive strategies.

Methods

Study population and design

We developed and validated a brain age estimation model using T1-weighted MRI scans from the UK Biobank[31] (UKB; application ID 89757), and evaluated its generalizability in two independent datasets: the Alzheimer's Disease Neuroimaging Initiative[32] (ADNI; https://adni.loni.usc.edu) and the Parkinson's Progression Markers Initiative[33] (PPMI; https://www.ppmi-info.org).

All datasets were de-identified and accessed under existing ethical approvals and data use agreements. UKB access was approved by the UK Biobank Access Committee and the North West Multi-centre Research Ethics Committee (REC06/MRE08/65), with written informed consent from participants. ADNI data were accessed via the LONI portal under the ADNI Data Use Agreement, with collection approved by local IRBs coordinated by the University of Southern California (USC), and informed consent allowing data sharing and secondary use. PPMI data were obtained under its Data Use Agreement and collected with Western IRB approval (Protocol 20131737), with consent covering data reuse. No additional IRB approval was required for this secondary analysis.

In the primary UKB cohort, we initially included 42,904 participants with baseline T1-weighted MRI scans. After filtering for genetically confirmed White British individuals aged 45–82 years ($n = 38{,}967$; 52.5% female) to minimize population stratification, we categorized participants into three groups: healthy controls ($n = 6036$) with no neurological or psychiatric diagnoses; a non-brain disease group ($n = 25{,}416$) with non-neurological comorbidities; and a brain disorder group ($n = 7515$) defined by ICD-10 codes for brain injury, neurodegeneration, or psychiatric conditions (Supplementary Table 2).

The ADNI cohort is a longitudinal, multicenter study including 1402 participants at baseline (cognitively normal [CN] = 332; mild cognitive impairment [MCI] = 785; Alzheimer's disease [AD] = 285; age 55–96 years; 44.0% male). Across follow-up, ADNI contributed 5058 T1-weighted scans: 1480 from CN participants (29.3%), 2836 from MCI (56.1%), and 742 from AD (14.7%) (Supplementary Table 3). Diagnoses were made using NIA-AA criteria, and data were harmonized across ADNI phases to control for scanner variability.

The PPMI cohort enrolled 1182 participants at baseline (CN = 185; prodromal Parkinson's disease=381; Parkinson's disease [PD]=616; age 45–83 years; 42.0% male). Over follow-up, 2907 T1-weighted scans were collected: 279 from CN (9.6%), 1326 from prodromal individuals (45.6%), and 1303 from PD patients (44.8%) (Supplementary Table 4). Diagnoses followed Movement Disorder Society (MDS) criteria, and MRI data were harmonized across 50 collection sites using a standardized preprocessing pipeline.

MRI data preprocessing

All MRI scans utilized in this study were acquired using 3 T MRI scanners. For the UK Biobank cohort, T1-weighted (T1w) MRI data were obtained in their publicly available preprocessed form. These images had undergone validated affine registration to the MNI152[34,35] standard space, resulting in an isotropic voxel resolution of 1 mm^3. For the external validation cohorts (ADNI and PPMI), raw T1w MRI images were processed using a standardized pipeline implemented in the FMRIB Software Library (FSL, version 6.0.5). To ensure methodological consistency, the preprocessing steps closely followed the UKB protocol. Specifically, raw images were first reoriented to match FSL's standard anatomical orientation and then cropped to remove non-brain regions, including the neck and inferior cranial areas. Bias field correction was applied without prior segmentation to address intensity non-uniformities while preserving anatomical fidelity. Non-brain tissues were subsequently removed using FSL's Brain Extraction Tool (BET). The images were then aligned to the MNI152 standard space using both linear and nonlinear transformations with six degrees of freedom. Finally, all images were uniformly resampled to dimensions of $182 \times 218 \times 182$ voxels, yielding a consistent isotropic spatial resolution of 1 mm^3. This harmonized preprocessing pipeline ensured standardization and compatibility across cohorts, providing a robust foundation for model training and evaluation.

Brain age estimation

Data preprocessing and model architecture. In this study, we trained a 3D Vision Transformer (3D-ViT) model using MRI data from the UKB to estimate brain age (Fig. 1a, Supplementary Table 5). Vision Transformer (ViT) models[29,36–39] have demonstrated superior performance over traditional machine learning and CNN-based architectures in various medical image analysis tasks. Thus, we chose the ViT framework for brain age gap (BAG) estimation over conventional CNN-based models. Briefly, T1-weighted MR images were resized to a resolution of $128 \times 128 \times 128$ as input to the model. The multi-head attention mechanism of the 3D-ViT divides each MRI scan into 3D patches (patch size: $16 \times 16 \times 16$). These patches are then processed through a transformer encoder, which computes attention scores across the patches. Finally, a multilayer perceptron head aggregates the transformer encoder outputs to generate the final brain age estimate.

Training, testing, and validation strategy

We evaluated the performance of the 3D-ViT model across both healthy and disease cohorts to assess its ability to detect brain age acceleration associated with various neurological and psychiatric conditions. The model demonstrated robust performance across multiple cohorts and subgroups, including both the UK Biobank (UKB) and external validation datasets (ADNI and PPMI). We followed a structured, multi-phase strategy for data

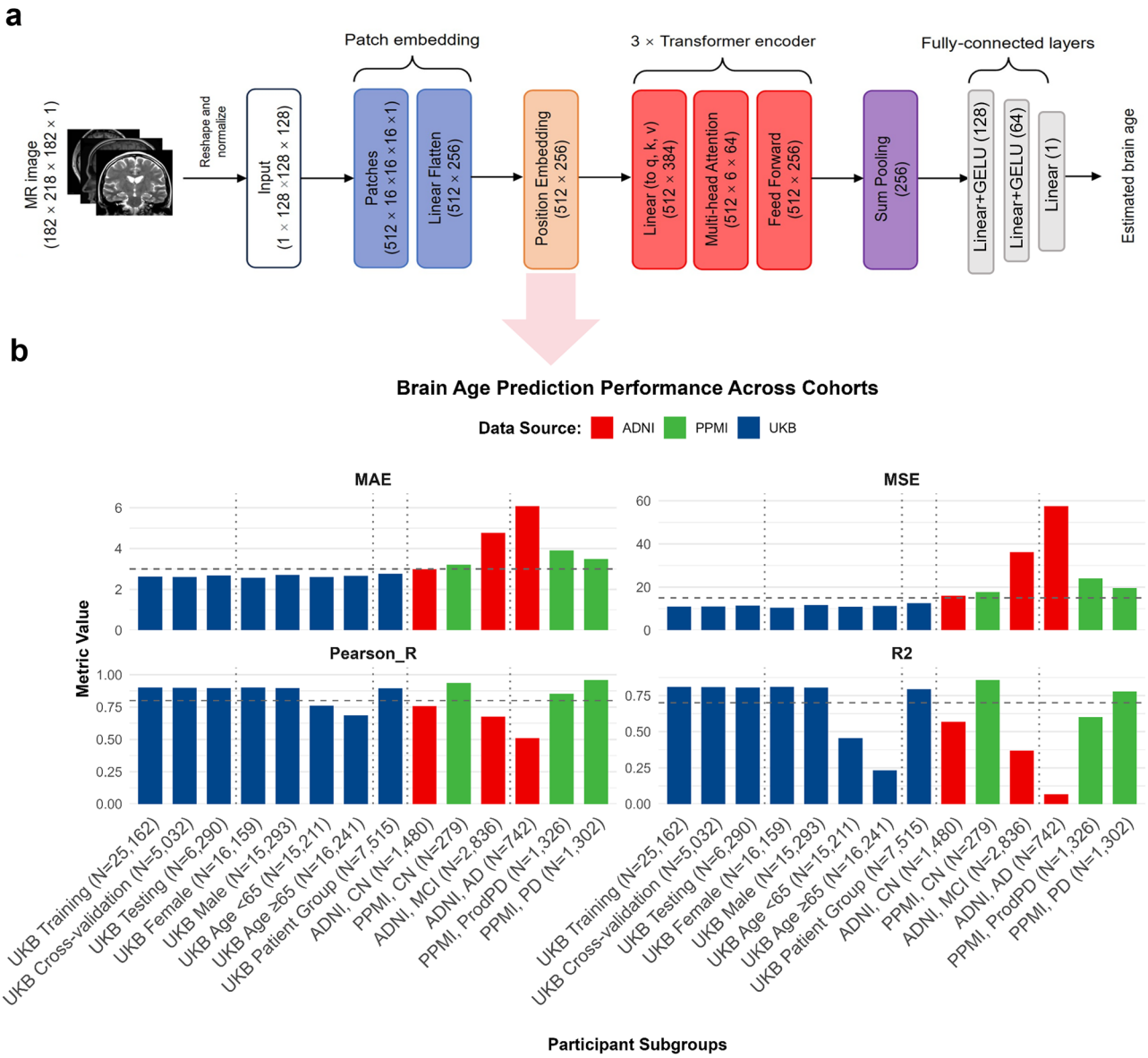

Fig. 1 | Comparative performance analysis of the 3D-ViT neural network in brain age estimation. a Architectural overview of the 3D-ViT model. **b** Performance metrics across demographic and clinical subgroups. Reported metrics include mean absolute error (MAE), mean squared error (MSE), coefficient of determination (R^2), and Pearson's correlation coefficient (r).

processing and analysis, as detailed in the data processing flowchart (Supplementary Figs. 1 and 2).

Model development and internal testing. We used data from the UKB, consisting of 31,452 participants without diagnosed brain disorders, and divided it into two groups: 6036 healthy controls (CN) with no documented neurological or psychiatric conditions, and 25,416 individuals with non-neurological comorbidities. We randomly split the dataset into a training set (80%; $n = 25{,}162$) and an internal testing set (20%; $n = 6290$). We employed the training set for model development, using a five-fold cross-validation approach to optimize model parameters and validate predictive performance. We assessed model performance using the mean absolute error (MAE), mean squared error (MSE), coefficient of determination (R^2), and Pearson's correlation coefficient (r) between the estimated brain age and chronological age. We optimized the model with the Adam optimizer over 200 epochs, using a learning rate of 2.0×10^{-5}. To prevent overfitting, we applied early stopping if the validation error did not improve within 10 epochs.

External validation on cognitively normal controls. We conducted external validation to ensure the robustness and generalizability of the developed model, using two independent datasets comprising cognitively normal participants: the ADNI cohort, which included 332 participants contributing 1480 scans, and the PPMI cohort, which included 185 healthy controls contributing 279 scans. These datasets were not involved in any stage of model training or hyperparameter tuning.

Independent validation on clinical disease cohorts. We assessed the model's clinical applicability and specificity by conducting additional validations on distinct clinical cohorts. These cohorts included participants from the UKB diagnosed with brain injuries, neurodegenerative diseases, or neuropsychiatric disorders ($n = 7515$), as well as patients from the ADNI and PPMI cohorts. The ADNI disease cohort consisted of participants diagnosed with MCI ($n = 785$; 2836 scans) and AD ($n = 285$; 742 scans). The PPMI disease cohort included individuals in the prodromal stage of Parkinsonism ($n = 381$; 1,326 scans) and those diagnosed with Parkinson's disease (PD) ($n = 616$; 1,303 scans). This structured

approach enabled rigorous internal and external validation, strengthening confidence in the generalizability and clinical relevance of the predictive models.

Quartile-based stratification of BAG

We calculated the Brain Age Gap (BAG) as the difference between neuroimaging-derived brain age and chronological age, providing a biomarker for accelerated neurobiological aging. Elevated BAG values (positive deviations) reflect disproportionately rapid brain aging relative to biological age, while negative values suggest maintenance patterns with preserved neuroanatomical integrity40. To improve clinical utility in predicting age-related disease trajectories, we stratified participants into quartiles based on the population-level BAG distribution: Q1 (slowest aging, 1st quartile), Q2 (2nd quartile), Q3 (3rd quartile), and Q4 (fastest aging, 4th quartile). This risk stratification framework enhances prognostic precision for neuropsychiatric disorders and supports targeted prioritization of high-risk subgroups (Q4) in population health strategies.

Association between BAG and cognitive performance

The association between brain age gap (BAG) and cognitive decline was examined in a broader UK Biobank (UKB) population ($N = 38{,}967$), which included both individuals without brain disease ($N = 25{,}416$) and those with diagnosed brain disorders ($N = 7515$). Of these, 25,617 participants with complete cognitive assessment records were included in the BAG–cognition association analysis (mean age = 64.5 ± 7.6 years; 52.3% female). Multi-domain cognitive performance was evaluated across seven core domains[40,41]—short-term memory, visual reasoning, abstract reasoning, processing speed, associative learning, executive function, and visual memory. Specific cognitive measures included the Numeric Memory Test (NMT) for digit span recall (range: 0–12); the Matrix Pattern Completion Test (MPCT) for visual pattern accuracy (range: 0–15); the Fluid Intelligence Test (FIT), consisting of 13 items assessing abstract reasoning; the Reaction Time Test (RTT), which measured median response latency in milliseconds; the Paired Associate Learning Test (PALT) for recall of word–image pairs (range: 0–10); the Tower Rearrangement Test (TR) for evaluating the number of steps to reach optimal puzzle solutions (range: 3–24); the Symbol Digit Substitution Test (SDST) for correct symbol–digit matches per minute; and the Pairs Matching Test (PMT) for visual recognition accuracy (range: 0–6).

Sample sizes varied slightly across cognitive assessments ($n = 24{,}872$–$25{,}617$) due to incomplete responses. Missing data were initially handled using listwise deletion to ensure analytical rigor (missingness per test ranged from 2.9% to 5.2%). To evaluate the potential impact of missingness, we conducted multiple imputation using chained equations[42] (MICE, 20 iterations), confirming that findings were consistent across imputed and complete-case datasets. Linear regression models were used to examine the association between BAG and cognitive performance across eight domains. In these models, the cognitive test scores served as the dependent variables, while BAG quartiles (Q1 as the reference group, reflecting the slowest aging) were the main independent variable. All models were adjusted for key covariates, including chronological age, sex, body mass index (BMI), smoking status, alcohol consumption. β coefficients and 95% confidence intervals (CIs) were estimated using ordinary least squares regression. To account for multiple testing across cognitive domains, we applied the Benjamini–Hochberg procedure to control the false discovery rate (FDR). To characterize nonlinear relationships between BAG and cognitive outcomes, we implemented restricted cubic spline (RCS) regression[43,44]. The number and position of knots were optimized using the Akaike Information Criterion (AIC)[45,46] to minimize overfitting while preserving model flexibility. Inflection points of the spline curves were extracted to identify threshold effects of BAG on cognition, and maximum effect sizes were reported for each domain. Statistical significance was assessed using two-sided P values with a significance threshold of $P < 0.05$. All continuous effect sizes were standardized (z-transformed) to enable cross-domain comparisons.

BAG as a risk marker for neurodegenerative, psychiatric, and neurological disorders

Case definitions and cohort stratification. We categorized neuropsychiatric disorders into three mechanistic groups based on shared pathophysiological features (Supplementary Tables 2–4): (1) Neurodegenerative disorders, including Alzheimer's disease (AD, ICD-10: G30), mild cognitive impairment (MCI)vascular dementia (VaD, F01), and Parkinson's disease (PD, G20); (2) Psychiatric disorders, including major depressive disorder (MDD, F32), anxiety disorders (F41), bipolar disorder (BD, F31), anorexia nervosa (AN, F50.0), obsessive-compulsive disorder (OCD, F42), post-traumatic stress disorder (PTSD, F43.1), and schizophrenia (SCZ, F20); and (3) Neurological disorders, including stroke (I63), epilepsy (G40), multiple sclerosis (MS, G35), and sleep disorders (SD, G47). Case status was determined using International Classification of Diseases, 10th Revision (ICD-10) codes extracted from hospital episode statistics. In the UK Biobank brain disorder cohort ($N = 7515$), the prevalence of disorders ranged from 26 (SCZ) to 4490 (MDD).

Survival analysis and follow-up parameters

Time-to-event analyses were performed using Cox proportional hazards models[47,48] to examine the longitudinal association between BAG and the incidence of neurological and psychiatric disorders. The primary exposure was defined as a 1-year increase in BAG, modeled as a continuous variable. In secondary analyses, BAG was stratified into quartiles (Q1 to Q4), with Q1 serving as the reference category. All models were adjusted for chronological age, sex, BMI, smoking status, and alcohol consumption to control for potential confounding.

Follow-up durations and censoring events were stratified by BAG quartiles and disorder type across three large cohorts. In the UK Biobank, median follow-up ranged from 9.00 to 9.07 years (maximum 26.97 years), with total person-years exceeding 360,000. The ADNI cohort was followed for up to 10.5 years (median = 2.0), with MCI and AD modeled as distinct outcomes. In the PPMI cohort, a fixed follow-up interval (median = 2 years) was applied for prodromal and PD cases. Cohort-level incidence data, person-year accumulation, and follow-up distributions by BAG quartile are detailed in Supplementary Tables 6–8 (Supplementary Table 6 is provided as Supplementary Data 8).

Hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated using Cox models for each disorder, with BAG quartiles as the main exposure. To address the issue of multiple testing across multiple disease outcomes, we applied the Benjamini–Hochberg procedure[49] to control the false discovery rate (FDR). Additionally, mean BAG values were compared between diagnostic groups and cognitively normal controls using two-sample t-tests.

Survival analysis of BAG and mortality risk

Survival analysis was conducted using Cox proportional hazards regression models, with chronological age specified as the underlying time scale. The analytic cohort included 38,967 UKB participants with complete mortality linkage through national death registries. The primary exposure variable was BAG, modeled both as a continuous predictor (per 1-year increase) and as a categorical variable stratified into quartiles (Q1 to Q4). The Q1 served as the reference category in all categorical models. All Cox models were adjusted for sex, BMI, smoking status, and alcohol consumption.

Participants were right-censored at the earliest of: (1) end of follow-up with confirmed survival, (2) loss to follow-up (<0.1% of the cohort), or (3) administrative study termination. The median follow-up time was 3.4 years (interquartile range [IQR]: 1.8–5.1 years), yielding a total of 148,774 person-years of observation. Censoring distributions and death rates across BAG quartiles were documented to assess exposure-outcome gradients (Supplementary Tables 9 and 10). To test the proportional hazards assumption, we evaluated both global and covariate-specific Schoenfeld residuals. No violations were detected (all p-values > 0.05; see Supplementary Table 11),

Table 1 | Seven modifiable lifestyle factors (UK Biobank) scored 0-1 per domain

Factor	Healthy Criteria	Assessment Tool
Smoking	Never smoker	NHS/NICE 2015 guidelines
Physical activity	≥150 min/week moderate or ≥75 min vigorous	IPAQ-SF
Alcohol	≤14 g/day (women), ≤28 g/day (men)	U.S. Dietary Guidelines
Diet	≥4/7 healthy food groups†	24 h dietary recall
Sleep	7–9 h/night	AASM/SRS consensus
Sedentary time	≤4 h/day non-work screen time	GPAQ
Social connection	Social isolation index ≤1‡	Social network survey

Healthy criterion denotes the predefined threshold for each behavior. Assessment tool denotes the validated questionnaire or guideline referenced.
†Fruits, vegetables, fish, whole grains, unprocessed meats, Low-fat dairy products, Nuts.
‡Composite of household size, social visit frequency, activity participation.

suggesting that no time-dependent covariates were incorporated into the final model.

In addition to the standard Cox models, we used RCS to model potential non-linear associations between BAG and mortality. Spline knots were optimized based on AIC, with one internal knot yielding the best fit. Non-linearity was assessed using a likelihood ratio test comparing the spline-based model to a linear model. All hazard ratios (HRs) and corresponding 95% confidence intervals (CIs) were estimated from the Cox models, and visualizations were generated to depict survival probabilities and HR functions across the full range of BAG values.

Lifestyle-BAG interaction analysis

Composite lifestyle score development. We constructed a 7-domain lifestyle score[50–53] for 38,967 participants using validated criteria (Table 1).

BAG interaction analysis

Each domain was scored as 1 (healthy) or 0 (unhealthy) based on established international guidelines and validated self-report tools. The total score ranged from 0 to 7, reflecting cumulative adherence to healthy lifestyle behaviors. Participants were categorized into three lifestyle classes: unfavorable (0–1), intermediate (2–4), and favorable (5–7). Participants were stratified into BAG quartiles (Q1–Q4). Within each quartile, lifestyle–BAG interactions were assessed in two steps. First, independent Student's t-tests were used to compare mean BAG values between individuals meeting versus not meeting healthy criteria for each lifestyle domain. Second, for categorical lifestyle class comparisons across BAG quartiles, the Kruskal–Wallis test[54] was applied to detect group-level differences, followed by Dunn's post-hoc test to adjust for multiple comparisons. All analyses were conducted using two-sided tests, with p-values < 0.05 considered statistically significant. BAG was treated as a continuous outcome variable, and data were reported as mean ± standard deviation (SD).

Statistics and reproducibility. All analyses were conducted in Python (v3.9) and R (v4.2.0), reporting continuous variables as mean ± SD or median (IQR) and categorical variables as counts (%). Brain age model performance was evaluated using MAE, MSE, R^2, and Pearson's r, with internal validation (5-fold cross-validation, 80/20 split) and external validation (ADNI/PPMI). Associations were tested via linear regression (adjusted for age/sex/BMI/smoking/alcohol) and Cox models (HRs with 95% CIs), with non-linear effects modeled using RCS and multiple testing controlled via FDR ($q < 0.05$). For reproducibility, UK Biobank (ID 89757), ADNI and PPMI data are publicly available, while preprocessing (FSL) and 3D-ViT scripts will be shared; all MRIs underwent identical preprocessing (MNI152, 1 mm^3), with fixed hyperparameters (Adam: lr = 2×10^{-5}, 200 epochs) and early stopping ensuring robustness. Missing data analyses (complete-case/MICE) yielded concordant results, with significance set at $p < 0.05$ (two-tailed).

Results

Performance of 3D-ViT in brain age estimation

Overall model performance. The 3D-ViT model was trained on MRI data from 31,452 UK Biobank participants, with 25,162 used for training (80%) and 6,290 reserved for testing (20%). During training, the model achieved robust performance (MAE = 2.627 years, MSE = 10.988, R^2 = 0.810, $r = 0.901$; (Fig. 1, Supplementary Fig. 3 and Supplementary Table 12). Five-fold cross-validation on 5,032 participants yielded an average MAE of 2.610 years, indicating strong model stability. On the independent test set ($N = 6290$), the model maintained consistent performance (MAE = 2.684 years, R^2 = 0.805, $r = 0.898$), demonstrating good generalizability. Stratified analyses showed similar performance by sex: females ($N = 16{,}159$) had an MAE of 2.567 years ($r = 0.902$) and males ($N = 15{,}293$) had an MAE of 2.714 years (r = 0.897), with a minimal difference of less than 0.2 years (Supplementary Fig. 4). Age-stratified analysis revealed a slightly higher MAE for participants aged ≥65 years ($N = 16{,}241$; MAE = 2.666 years, $r = 0.687$) compared to those <65 years ($N = 15{,}211$; MAE = 2.609 years, $r = 0.760$; Supplementary Fig. 4). Further analyses showed that females and younger participants had lower mean BAGs than males and older individuals (Supplementary Fig. 5). Both underweight and overweight individuals exhibited higher BAGs compared to those with normal BMI, highlighting the adverse impact of abnormal BMI on brain health.

External validation. External validation was performed using two independent datasets: ADNI (1,480 scans, including 332 cognitively normal subjects) and PPMI (279 scans, including 185 healthy controls). The model achieved an MAE of 2.998 years (MSE = 16.000, $r = 0.758$) in ADNI (Supplementary Fig. 6), and an MAE of 3.205 years (MSE = 17.640, $r = 0.936$) in PPMI (Supplementary Fig. 7). The slightly higher error in ADNI ($\Delta \approx 0.3$ years compared to UKB) is likely due to the older baseline age of the cohort. In contrast, PPMI's higher correlation might be attributed to the narrower age distribution within the dataset. Despite these cohort-specific differences, the model's consistent performance across these external datasets further supports its cross-dataset applicability and robustness.

Disease-associated brain age acceleration

The 3D-ViT model accurately estimated brain age across healthy and diseased populations, capturing marked disease-related brain age acceleration, particularly in neurodegenerative and psychiatric disorders. In the ADNI dataset, which includes individuals with MCI and AD, the model demonstrated notable brain age acceleration (Supplementary Fig. 6). The MCI group ($N = 2836$) showed a MAE of 4.77 years and a correlation of $r = 0.675$. In contrast, AD patients ($N = 742$) exhibited a further increase in MAE to 6.08 years, with a lower correlation of $r = 0.51$. In the PPMI dataset, PD patients ($N = 1302$) displayed moderate brain age acceleration (MAE = 3.48 years, $r = 0.96$), while prodromal PD patients ($N = 1326$) had an MAE of

3.91 years ($r = 0.854$), indicating that the model can detect early neurodegenerative changes (Supplementary Fig. 7). In the UK Biobank, the healthy control group showed an MAE of 2.684 years, whereas individuals with psychiatric disorders exhibited higher MAEs (Supplementary Fig. 8). Specifically, individuals with MDD ($N = 4{,}490$) exhibited an MAE of 2.718 years ($r = 0.896$). ANX ($N = 2390$) and BD ($N = 72$) had MAEs of 2.697 and 3.291 years, respectively, indicating measurable brain aging in these cohorts. In neurological disorders, the stroke group ($N = 743$) exhibited an MAE of 2.869 years ($r = 0.866$), while epilepsy patients ($N = 340$) had an MAE of 2.910 years ($r = 0.898$). The MS group ($N = 151$) showed the highest MAE at 5.131 years ($r = 0.724$), suggesting relatively more accelerated brain aging in this cohort. These findings demonstrate that the 3D-ViT model effectively captures disease-associated brain age acceleration, with performance varying by disease severity and type.

Association between BAG and cognitive decline

Cognitive performance was stratified by BAG quartiles in the UK Biobank cohort ($n = 25{,}617$, Supplementary Tables 13 and 14, Supplementary Fig. 9). Significant differences were observed across quartiles for several cognitive domains. Matrix pattern completion scores were higher in Q1 (8.10 ± 2.10) and Q2 (8.12 ± 2.07) compared to Q3 (7.95 ± 2.11) and Q4 (7.95 ± 2.12) ($F = 12.40$, $p < 0.001$). Reaction time was significantly slower in Q4 (604.88 ± 112.78 ms) compared to Q1 (591.74 ± 106.26 ms) ($F = 19.27$, $p < 0.001$). Paired associate learning scores were lower in Q4 (6.85 ± 2.52) compared to Q1 (7.14 ± 2.52) ($F = 19.82$, $p < 0.001$). Symbol digit substitution scores were also lower in Q4 (18.45 ± 5.21) compared to Q1 (19.29 ± 5.15) ($F = 40.67$, $p < 0.001$). No significant differences were observed for numeric memory ($F = 0.52$, $p = 0.671$) or fluid intelligence ($F = 1.93$, $p = 0.122$).

Linear regression models revealed significant associations between biological aging quartiles and cognitive performance (Supplementary Table 15 and Supplementary Fig. 10). Compared to Q1, Q3 and Q4 showed significantly lower scores in matrix pattern completion (Q3: $\beta = -0.15$, $p < 0.001$; Q4: $\beta = -0.15$, $p < 0.001$), paired associate learning (Q3: $\beta = -0.14$, $p = 0.001$; Q4: $\beta = -0.28$, $p < 0.001$), and symbol digit substitution (Q3: $\beta = -0.52$, $p < 0.001$; Q4: $\beta = -0.84$, $p < 0.001$). Reaction time was significantly slower in Q3 ($\beta = 8.06$, $p < 0.001$) and Q4 ($\beta = 13.15$, $p < 0.001$) compared to Q1. No significant associations were found for numeric memory or fluid intelligence after adjusting for multiple comparisons using the Benjamini-Hochberg procedure.

Non-linear modeling identified inflection points and maximum effects of BAG on cognitive domains (Fig. 2 and Supplementary Table 16). The maximum effects across all domains were observed in Q4, indicating that the most pronounced cognitive deficits occurred in individuals with the highest BAG. For instance, reaction time exhibited an inflection point at BAG ≈ −1.75, with a maximum effect of 3.13 ms in Q4. Similarly, paired associate learning showed a maximum effect of 3.11 in Q4, with an inflection point at BAG ≈ 0.01. These results suggest that cognitive performance remains relatively stable at lower BAG levels but declines more sharply beyond a critical threshold, particularly in Q4. We conducted a detailed comparative analysis between imputed and unimputed datasets, restricted to individuals without missing cognitive values. The results confirmed that BAG-cognition associations and model-estimated inflection points remained consistent across datasets, with no meaningful differences observed (Supplementary Fig. 11 and Supplementary Table 17).

BAG as a risk marker for neuropsychiatric disorders

Cross-sectional analysis based on the BAG revealed significant heterogeneity in BAG levels between disease groups and the CN (BAG: 0.122 ± 3.039; Fig. 3b, Supplementary Table 18). Among neurodegenerative disorders, Alzheimer's disease (AD: 3.242 ± 6.635, $p < 0.001$) and patients with MCI showed significantly increased (BAG = 2.06 ± 5.62 years, $p < 0.001$), with subgroup analyses revealing progressive BAG elevation in early MCI (EMCI: 2.33 ± 5.51 years) and late MCI (LMCI 3.12 ± 5.68 years) stages (Supplementary Fig. 12; Supplementary Tables 19 and 20). Whereas VaD (0.279 ± 3.463, $p = 0.442$) and PD (0.003 ± 4.463, $p = 0.321$) showed no significant differences compared to CN. Notably, ProdPD (−1.441 ± 4.880, $p < 0.001$) demonstrated a unique negative BAG shift. In psychiatric disorders, BD (1.913 ± 4.051, $p < 0.001$), MDD (0.516 ± 3.409, $p < 0.001$), and ANX (0.525 ± 3.425, $p < 0.001$) displayed significantly higher BAG than CN, whereas OCD (1.161 ± 3.231, $p = 0.074$), PTSD (0.310 ± 3.539, $p = 0.645$), and AN (0.369 ± 3.547, $p = 0.631$) showed no significant differences. SCZ (2.068 ± 3.341, $p = 0.007$) exhibited elevated BAG despite a limited sample size ($n = 26$). Among neurological disorders, multiple sclerosis (4.069 ± 5.328, $p < 0.001$) showed the greatest elevation in BAG, while stroke (0.631 ± 3.585, $p < 0.001$), epilepsy (1.059 ± 3.478, $p < 0.001$), and sensory disorder (0.641 ± 3.379, $p < 0.001$) also exhibited higher brain age gaps compared to controls.

Our study investigated the association between BAG and the risk of 16 neuropsychiatric disorders (Fig. 3c, Supplementary Fig. 13, and Supplementary Table 21, provided as Supplementary Data 9), categorized into neurodegenerative, psychiatric, and neurological groups, using Cox regression models adjusted for chronological age, sex, BMI, smoking status, and alcohol consumption. Elevated BAG was strongly associated with neurodegenerative diseases, particularly AD and MCI, with each 1-year increase in BAG conferring a 16.5% higher risk of AD (HR = 1.165, 95% CI = 1.086–1.249, $p < 0.001$) and a 4.0% higher risk of MCI (HR = 1.040, 1.030–1.050, $p < 0.001$). Participants in the highest BAG quartile (Q4) had 2.8-fold increased AD risk (HR = 2.801, $p = 0.011$) and 1.7-fold increased MCI risk (HR = 1.691, $p < 0.001$) compared to Q1. VaD showed no significant BAG association (HR = 1.022, $p = 0.323$), aligning with its distinct cerebrovascular pathophysiology. PD exhibited mixed results: prodromal PD had reduced BAG (mean = −1.441 vs. controls, $p < 0.001$), while incident PD showed no significant association (HR = 1.038, $p = 0.385$), reflecting its focal nigrostriatal degeneration. In psychiatric disorders, BAG independently predicted MDD (HR = 1.046 per year, $p < 0.001$), BD (HR = 1.174, $p < 0.001$), and ANX (Q4 HR = 1.442, $p < 0.001$), with Q4 participants facing 1.5-fold higher MDD risk (HR = 1.466, $p < 0.001$) and 2.4-fold higher BD risk (HR = 2.431, $p = 0.038$) versus Q1. SCZ risk escalated sharply in Q4 (HR = 7.504, $p = 0.022$), albeit with limited case numbers ($n = 26$). Among neurological disorders, stroke risk increased by 5.5% per BAG year (HR = 1.055, $p < 0.001$), with Q4 participants facing 1.6-fold higher risk (HR = 1.612, $p < 0.001$), while multiple sclerosis (MS) demonstrated the most pronounced BAG association (Q4 HR = 6.417, $p < 0.001$), consistent with its inflammatory-driven accelerated aging[55]. Epilepsy risk doubled in Q4 (HR = 2.189, $p < 0.001$), likely reflecting cumulative neurotoxic effects.

Survival analysis of BAG and all-cause mortality

We conducted a comprehensive survival analysis to evaluate the association between BAG and all-cause mortality using data from national death registries linked to participant records in the UKB. The cohort was followed for a median of 3.4 years (range: 0–7.5 years), with a total follow-up time of 148,774 person-years. During the follow-up period, 469 participants (1.2%) died from various causes, while 38,498 participants (98.8%) were right-censored. Censoring rates exhibited a graded pattern across BAG quartiles: Q1 (99.2%), Q2 (99.1%), Q3 (98.8%), and Q4 (98.2%). Mortality rates increased progressively across BAG quartiles: Q1 (0.8%), Q2 (0.9%), Q3 (1.2%), and Q4 (1.8%). Cox regression models adjusted for chronological age, sex, BMI, smoking, and alcohol consumption revealed significant mortality associations. Per 1-year increase in BAG conferred a 12% elevated mortality risk (adjusted HR 1.12, 95% CI 1.09–1.15; $p < 0.001$). Quartile-based analysis demonstrated a dose-response relationship: compared with Q1 (reference), Q2 showed non-significant risk elevation (HR 1.18, 0.87–1.59; $p = 0.293$), while Q3 (HR 2.11, 1.19–2.59; $p = 0.002$) and Q4 (HR 2.36, 1.81–3.08; $p < 0.001$) exhibited progressively stronger associations. The proportional hazards assumption remained valid across all covariates (global Schoenfeld test $p = 0.88$; covariate-specific $p > 0.68$, Supplementary Tables 11). Kaplan-Meier survival curves confirmed significant interquartile divergence (log-rank $p < 0.001$). Q4 participants demonstrated

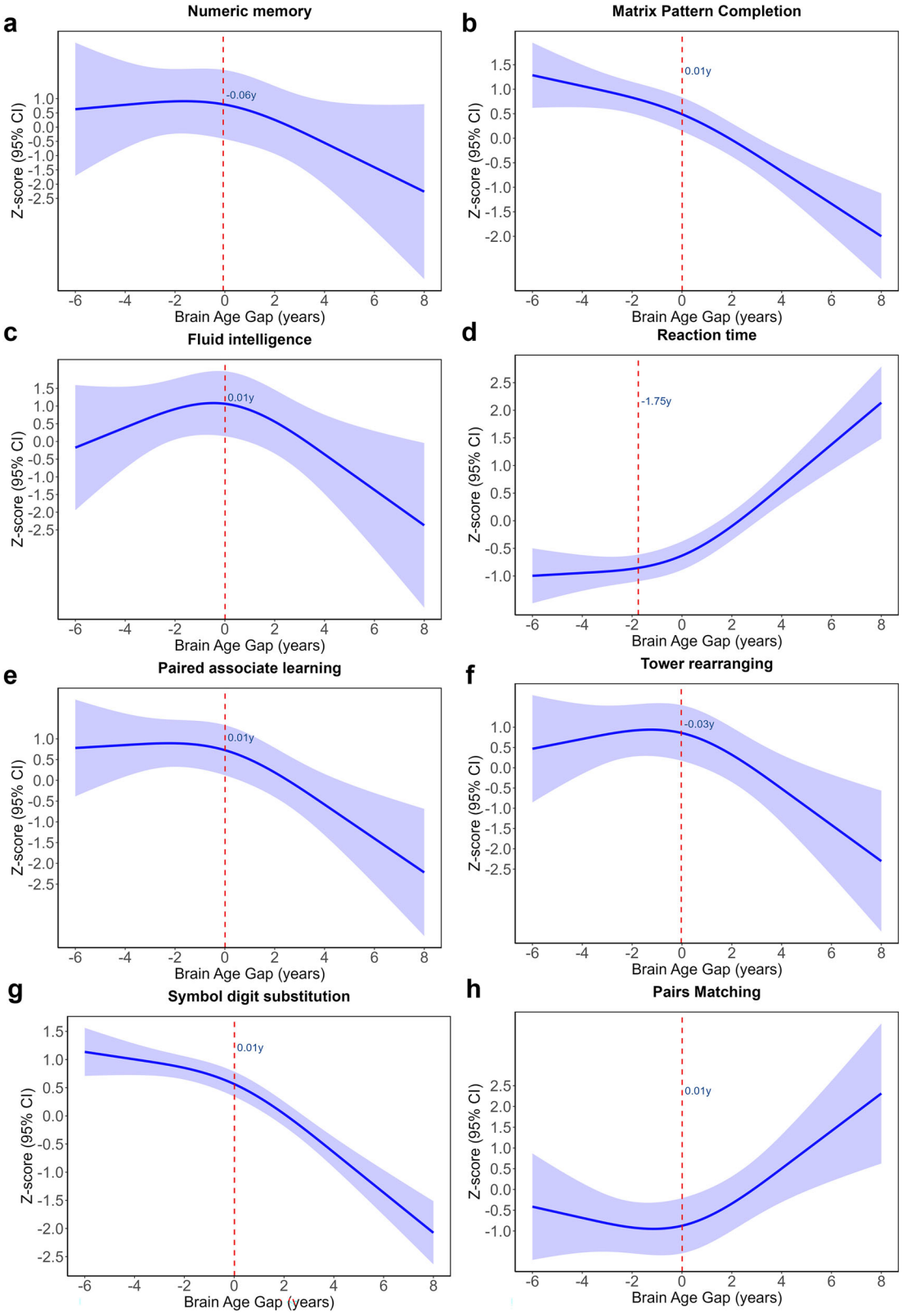

Fig. 2 | Dose-response relationships between BAG and cognitive performance across neuropsychological domains. Panels show restricted cubic spline (RCS) curves illustrating the non-linear associations between BAG and performance in eight cognitive domains (**a**: Numeric Memory; **b**: Matrix Pattern Completion; **c**: Fluid Intelligence; **d**: Reaction Time; **e**: Paired Associate Learning; **f**: Tower Rearranging; **g**: Symbol Digit Substitution; **h**: Pairs Matching). The models are adjusted for chronological age, sex, BMI, smoking status, and alcohol consumption. Solid blue lines represent estimated effects, with shaded bands indicating 95% confidence intervals. The vertical red dotted line indicates the inflection point in each spline—the BAG value at which the rate of change in cognitive performance is maximized.

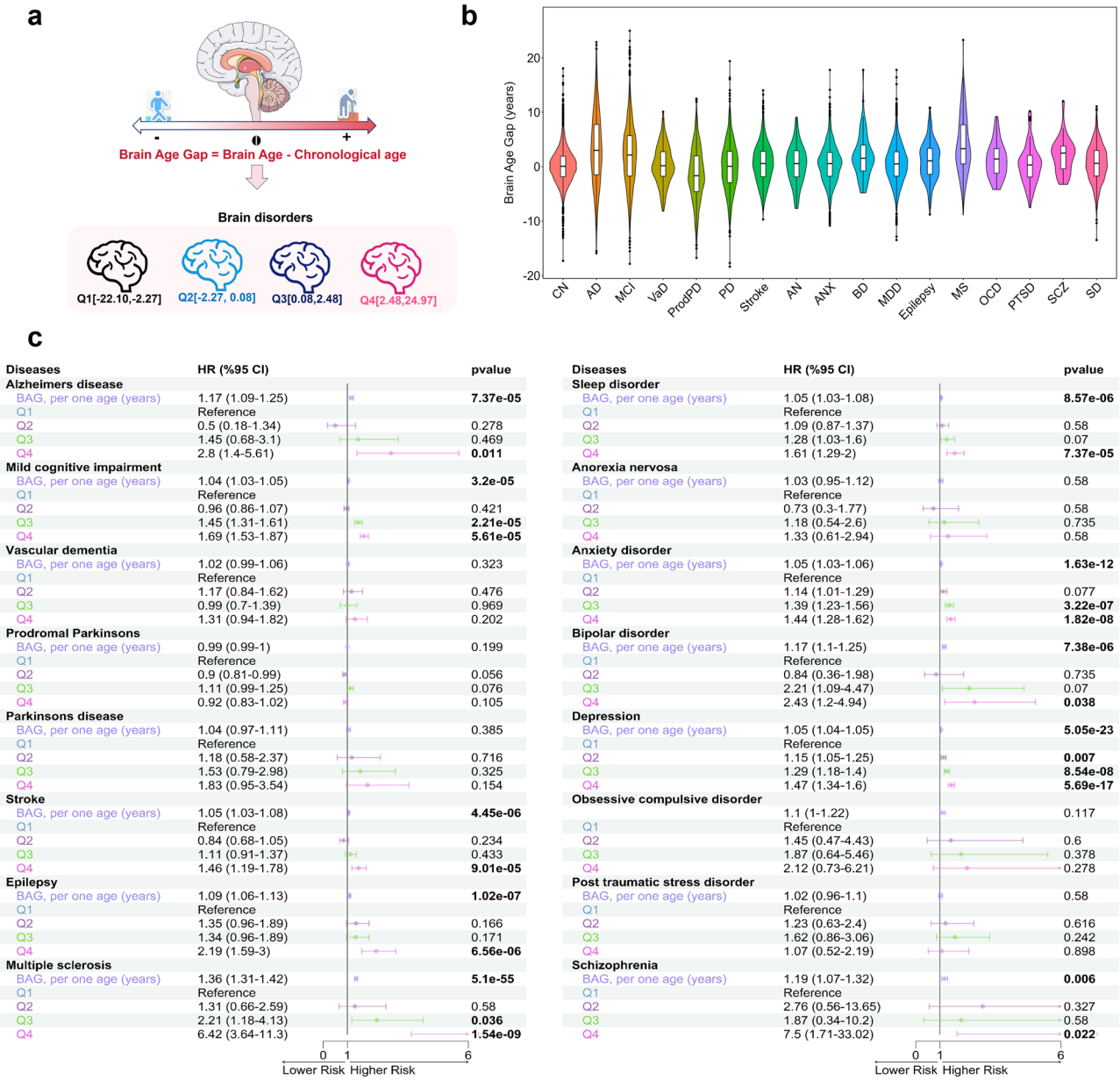

Fig. 3 | Evaluating the ability of BAG to assess neuropsychiatric disorders. **a** Conceptual framework of BAG and its stratification. BAG is defined as the difference between predicted brain age and chronological age, serving as a biomarker of accelerated brain aging. Individuals were stratified into quartiles (Q1–Q4) based on the BAG distribution, with Q4 representing the highest deviation from expected aging patterns. **b** Distribution of BAG across diagnostic groups. Violin plots illustrate BAG distributions in healthy controls (CN) and individuals diagnosed with Alzheimer's disease (AD), mild cognitive impairment (MCI), vascular dementia (VaD), prodromal Parkinson's disease (ProdPD), Parkinson's disease (PD), stroke, epilepsy, multiple sclerosis (MS), sleep disorder (SD), anorexia nervosa (AN), anxiety disorders (ANX), bipolar disorder (BD), major depressive disorder (MDD), obsessive-compulsive disorder (OCD), post-traumatic stress disorder (PTSD), and schizophrenia (SCZ). Wider sections of each plot indicate higher data density, with color coding corresponding to disease classification. **c** Associations between BAG and disease risk across the same diagnostic groups. Hazard ratios (HR) with 95% confidence intervals (CIs) are estimated from Cox proportional hazards models, evaluating risk both per unit increase in BAG and across BAG quartiles (Q1–Q4). Asterisks indicate statistical significance levels (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). Whiskers in panel b extend to 1.5 times the interquartile range (IQR) from the lower and upper quartiles. Whiskers in panel c represent 95% confidence intervals. Sample sizes and statistical comparisons are provided in Supplementary Tables 18 and 21.

substantially reduced survival probability versus Q1 (absolute difference 3.0% at 7 years), with intermediate quartiles showing graded risk progression (Fig. 4a). Nonlinear restricted cubic spline analysis demonstrated a robust and statistically significant association between BAG and mortality risk (Fig. 4b, likelihood ratio test for nonlinearity, $p < 0.001$). Mortality risk remained stable at BAG values below 2 years (HR ≈ 1.0) but increased sharply thereafter, with a HR of 1.2 at 3 years and exceeding 2.0 at BAG values of 6–8 years. These findings underscore the importance of monitoring and addressing elevated BAG levels to mitigate associated mortality risks.

Lifestyle-BAG interaction

Stratified analyses revealed a significant dose–response relationship between composite lifestyle scores and BAG progression across quartiles in the overall population ($N = 38{,}967$; $p < 0.001$ for trend; Fig. 5, Supplementary Tables 22 and 23). The most pronounced neuroprotective effects were

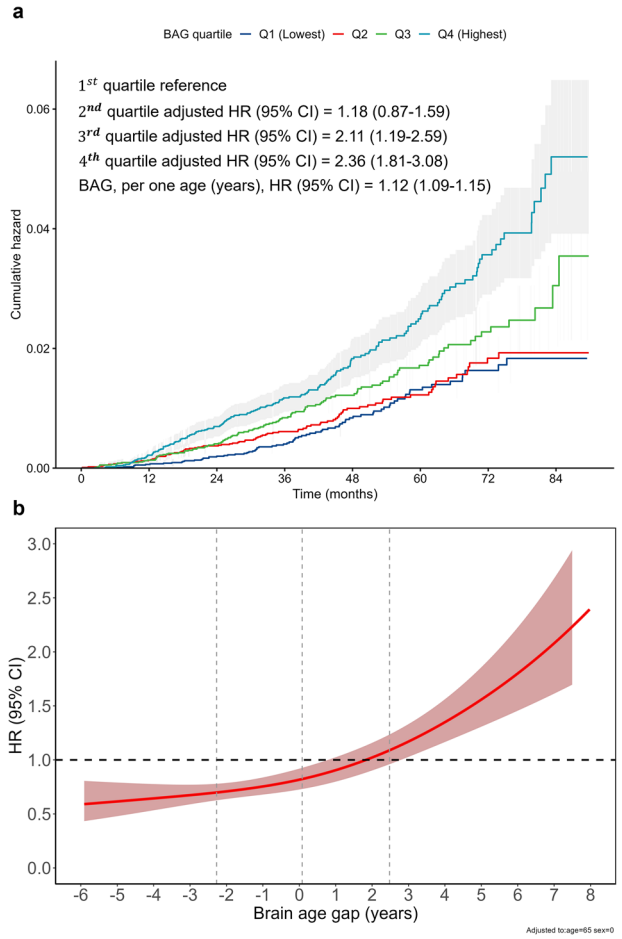

Fig. 4 | BAG predicts all-cause mortality via non-linear associations. **a** Kaplan–Meier survival curves stratified by BAG quartiles ($N = 38967$; two-sided log-rank test $p < 0.001$; adjusted for age, sex, BMI, smoking status and alcohol consumption). **b** Restricted cubic spline curves showing non-linear association between BAG and mortality risk (likelihood ratio test for non-linearity $p < 0.001$; single internal knot). Hazard ratios and 95% confidence intervals were estimated from Cox proportional hazards models incorporating RCS; the horizontal black dotted line at HR = 1 denotes the reference (no change in risk).

observed in Q4, where individuals with favorable lifestyles (scores 5–7) exhibited a 0.35-year lower BAG compared to those with unfavorable profiles (scores 0–1; 4.15 ± 1.63 vs. 4.50 ± 1.81; $p < 0.001$). Intermediate lifestyles (scores 2–4) conferred modest protection (1.7% annual reduction; $p = 0.013$). Domain-specific analysis in Q4 identified moderate alcohol consumption as the strongest protective factor (–0.20 years; $p < 0.001$), followed by regular physical activity (–0.14 years; $p = 0.026$), and never smoking (–0.11 years; $p < 0.001$). Stage-dependent patterns revealed transient sleep benefits restricted to Q1 (-0.09-year, $p = 0.026$). Threshold phenomena were observed for sedentary behavior (2.3% BAG reduction in Q1 [$p = 0.046$], nonsignificant in Q3–Q4), while dietary patterns showed no significant associations across strata ($p = 0.267$).

To reinforce these findings, subgroup analyses were conducted in participants without diagnosed brain disorders ($N = 31,452$; Supplementary Table 24) and in those with neurological and psychiatric conditions ($N = 7515$; Supplementary Table 25). In both the overall and non-brain disorder populations, lifestyle behaviors showed consistent and graded associations with BAG, particularly in Q4. In contrast, within the neurological and psychiatric disorder group, lifestyle behaviors did not exhibit significant associations with BAG or cognitive outcomes across quartiles, suggesting potential limits to behavioral modifiability in the presence of established brain pathology.

Discussion

Our study aimed to investigate the associations between BAG and key clinical outcomes within a large, population-based cohort. We identified robust relationships between elevated BAG and domain-specific cognitive decline, increased risks of neuropsychiatric disorders, and heightened all-cause mortality. Importantly, our findings suggest that adopting healthy lifestyle interventions may effectively mitigate the negative consequences associated with accelerated brain aging. To achieve precise estimation of brain age, we employed a 3D-ViT deep learning framework, which demonstrated strong potential in providing reliable and biologically meaningful brain aging biomarkers. Collectively, these results support the application of BAG as an effective tool for identifying individuals at high risk, guiding targeted clinical interventions, and informing public health strategies aimed at promoting brain health and longevity.

In our previous work[6], our comparative analyses of model architectures demonstrated that 3D-ViT consistently outperformed conventional CNNs and hybrid models (including VGGNet[56], ResNet[57], DenseNet[58], EfficientNet[59], Simple fully convolutional network (SFCN)[60], and Global-local transformer (GLT)[61] across four multicenter datasets (UKB, ADNI, PPMI, and IXI) (Supplementary Table 26). The principal advantage manifested in prediction accuracy: for the clinically most relevant metric of mean absolute error (MAE), 3D-ViT achieved 2.60 years in the UKB test set (0.04 years lower than the suboptimal DenseNet model), with external validation results of 2.99 years (ADNI) and 3.44 years (PPMI), surpassing all mainstream comparative models (e.g., ResNet's 3.17 years in ADNI and 4.03 years in PPMI). Notably, in the IXI dataset, 3D-ViT yielded an MAE of 3.61 years, significantly better than the optimal comparator ResNet (4.04 years), demonstrating robust cross-protocol adaptability. These findings collectively indicate that the Transformer-based architecture excels in capturing complex brain aging patterns, exhibiting superior predictive accuracy, robustness, and generalizability compared to conventional deep learning approaches.

We further confirmed the biological plausibility and clinical applicability of our model: (1) Large-scale training validation: Using the UKB dataset ($N = 31,452$), the model demonstrated stable performance in five-fold cross-validation (MAE = 2.610 ± 0.15 years, $R^2 = 0.812$) and maintained consistency in independent validation cohorts (ADNI: MAE = 3.205 years; PPMI: MAE = 2.998 years). (2) Multicenter robustness: The model exhibited unbiased estimation across the UKB test set ($N = 6290$), ADNI ($N = 1480$), and PPMI ($N = 279$) cohorts, with minimal performance variation between sexes ($|\Delta MAE| < 0.2$ years) and age groups (≥65 vs. <65 years: ΔMAE = 0.06 years; ANCOVA, $p > 0.1$). Transfer learning further confirmed the robustness of extracted neuroimaging features. (3) Pathological sensitivity: The 3D-ViT effectively quantified BAG associated with disease staging and severity. Notably, AD (BAG = 3.24 ± 6.64 years, $p < 0.001$) and MS (BAG = 4.07 ± 5.33 years, $p < 0.001$) patients showed significant BAG elevation (Supplementary Table 18). The 3D-ViT model demonstrates significant advantages over the hybrid approach combining ResNet-50 with kernel regression proposed by Pilli et al.[62–64]. While conventional kernel methods rely on handcrafted features and shallow networks, limiting their capacity to fully capture the complexity of brain aging, our model achieves sensitive detection of global-local atrophy patterns through end-to-end whole-brain MRI processing. In the ADNI cohort, it not only achieves comparable prediction accuracy (MAE = 2.99 years) but also robustly differentiates between CN (BAG = 0.66 ± 3.45 years), early MCI (EMCI, 2.33 ± 5.51 years), late MCI (LMCI, 3.12 ± 5.68 years), and AD (BAG = 3.53 ± 6.80 years) groups ($p < 0.01$), thereby providing a useful tool for monitoring AD progression (Supplementary Table 19). Additionally, elevated BAG in individuals with abnormal BMI (ΔMAE = 0.193–0.399 years) suggested metabolic-neurodegenerative interactions[65], broadening clinical applications. Intriguingly, MS patients in the UKB psychiatric cohort exhibited higher BAG (4.069 years) than AD patients (ΔBAG = 0.827 years), providing additional insights into the neurobiology of psychiatric disorders.

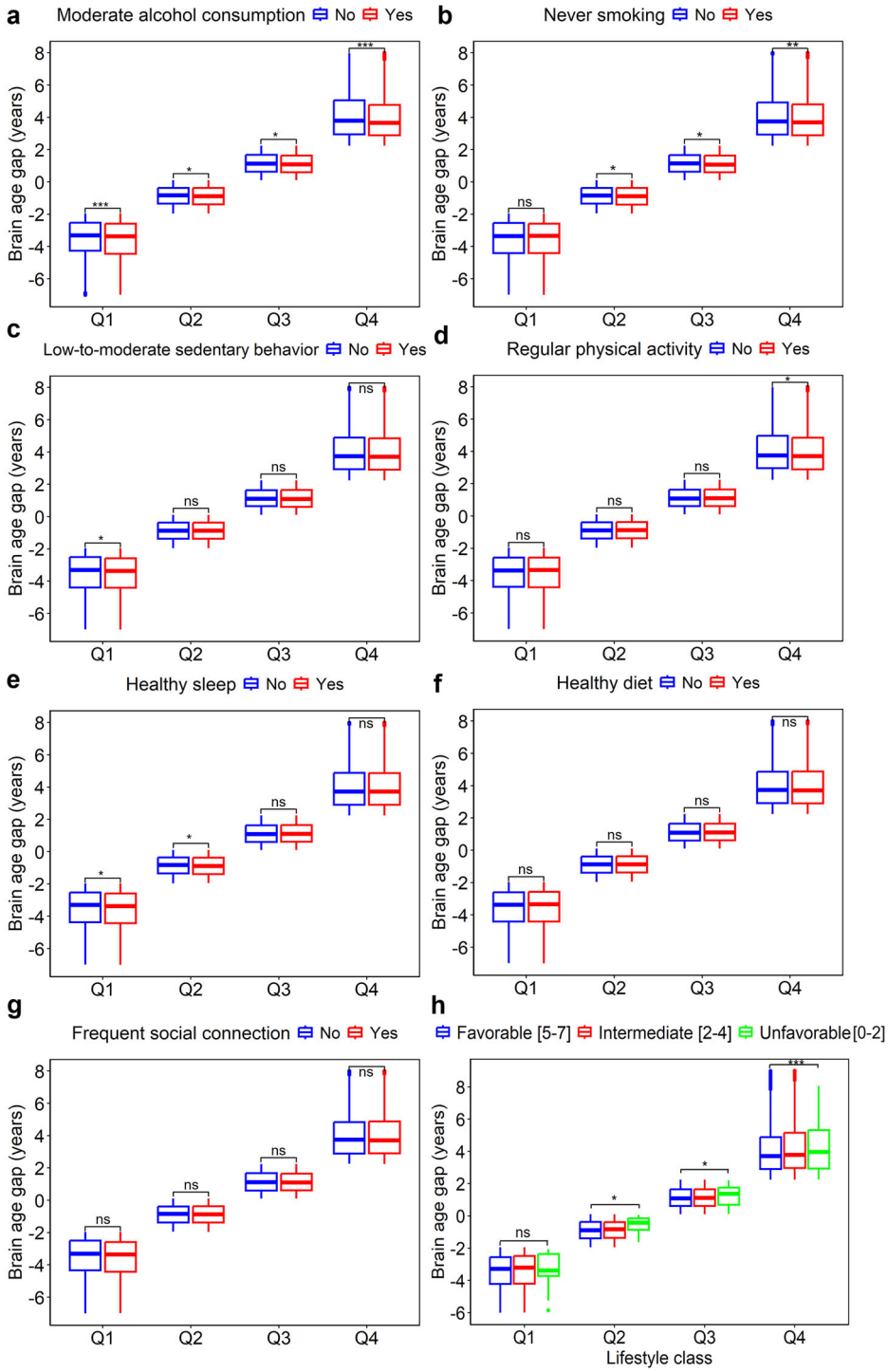

Fig. 5 | Associations between lifestyle factors and BAG across quartiles (Q1–Q4; total $N = 38{,}967$). Data are stratified by absence ("No", blue) or presence ("Yes", red) of each healthy behaviour. Exact Pvalues for the "No" vs "Yes" comparison are shown for moderate alcohol consumption (**a**: Q14.82 × 10^{-4}; Q20.036; Q30.040; Q44.63×10^{-7}), never smoking (**b**: 0.427; 0.045; 0.043; 4.60×10^{-3}), low-to-moderate sedentary behaviour (**c**: 0.046; 0.950; 0.565; 0.353), regular physical activity (**d**: 0.620; 0.357; 0.580; 0.026), healthy sleep (**e**: 0.026; 0.019; 0.670; 0.352), healthy diet (**f**: 0.769; 0.620; 0.409; 0.267), frequent social connection (**g**: 0.055; 0.755; 0.317; 0.661) and overall lifestyle classification (favourable5–7 vs unfavourable0–1; **h**: Q10.180; Q20.013; Q30.012; Q42.35 × 10^{-5}). Significance is denoted by $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$; ns, not significant. Whiskers extend to 1.5× interquartile range. Detailed subgroup sample sizes are provided in Supplementary Tables 22 and 23.

Our study provides robust evidence of a significant association between the BAG and domain-specific cognitive decline in the UKB cohort. These findings enhance our understanding of neurobiological aging by identifying critical thresholds and nonlinear relationships between structural brain changes and cognitive performance. Stratification by BAG quartiles revealed a clear gradient in cognitive outcomes, with the most pronounced deficits observed in the highest quartile (Q4; BAG > 2.48 years). Individuals in Q4 exhibited significantly slower reaction times, reduced executive function (symbol digit substitution: $\beta = -0.84$ vs. Q1), and impaired associative memory (paired associate learning: $\beta = -0.28$ vs. Q1). These results align with prior neuroimaging studies[66–68] implicating frontostriatal and hippocampal networks in age-related cognitive decline. Importantly, our analysis indicates distinct cognitive trajectories across BAG quartiles. Accelerated brain aging correlates with measurable cognitive impairment even among apparently healthy middle-aged and older adults. Previous studies[69–72] have reported similar associations, demonstrating that higher BAG is linked to poorer cognitive and motor functions in populations such as Parkinson's disease patients and older adults with decreased processing speed and memory performance. Identifying domain-specific vulnerabilities through BAG highlights its potential as a biomarker for early detection of subclinical cognitive decline. Individuals surpassing a critical BAG threshold (>2.48 years) appear especially vulnerable to substantial

cognitive deterioration, suggesting a critical intervention window for targeted preventive strategies.

Our large-scale study demonstrates that the BAG exhibits distinct pathophysiological associations across neurodegenerative, psychiatric, and neurological disorders, while also reflecting shared mechanisms underlying neuropsychiatric conditions. The robust association between elevated BAG and AD (HR = 1.165/year, $p < 0.001$) is consistent with prior neuroimaging evidence showing widespread cortical thinning and structural brain changes in Alzheimer's disease[73]. Intriguingly, the paradoxical reduction in BAG observed in prodromal Parkinson's disease (Q4 HR = 0.925, $p = 0.105$) may reflect compensatory neuroplasticity[74,75] during early nigrostriatal degeneration, whereas the absence of BAG effects (Q4 HR = 1.830, $p = 0.154$) in incident PD suggests that focal dopaminergic loss minimally impacts global brain aging metrics[76]. Clinically, PD is characterized by substantia nigra dopaminergic neuron degeneration. However, our brain age model utilizes whole-brain volumetric indices, which may lack sensitivity to PD-specific neurodegeneration localized in basal ganglia-thalamocortical circuits—regions that minimally affect global brain volume[6,39]. The dose-dependent relationship between BAG and psychiatric disorders further highlights its transdiagnostic utility. For MDD, the quartile-dependent risk escalation (Q4 HR = 1.466, $p < 0.001$) supports the hypothesis of accelerated aging mediated by chronic stress-induced glucocorticoid toxicity and mitochondrial dysfunction[77,78]. Notably, SCZ exhibited an exponential risk surge in the highest BAG quartile (HR = 7.504, $p = 0.022$), suggesting that neurodevelopmental deficits in synaptic pruning may synergize with aging-related cortical thinning to amplify vulnerability[79,80].

In neurological disorders, the pronounced association between BAG and MS (Q4 HR = 6.417, $p < 0.001$) points to inflammaging—a convergence of chronic neuroinflammation and epigenetic aging—as a driver of disability progression[81,82]. Similarly, the 5.5% annual increase in stroke risk per BAG year ($p < 0.001$) implicates endothelial glycocalyx degradation and blood-brain barrier disruption as mediators of neurovascular aging[83,84]. Current evidence[85,86] demonstrates that patients with epilepsy (including temporal lobe epilepsy, idiopathic generalized epilepsy, and progressive myoclonus epilepsy subtypes) exhibit significantly accelerated brain aging compared to healthy populations, which aligns with the findings of this study. This abnormal brain-age advancement may stem from neuropathological damage induced by recurrent seizures. Notably, however, premature brain aging persists even in well-controlled epilepsy types (e.g., idiopathic generalized epilepsy), suggesting that inherent susceptibility factors or interictal epileptiform discharges may independently contribute to the cerebral aging process. These findings support the concept of brain age as a comprehensive indicator of cerebral health: when the estimated brain age exceeds the chronological age, it often suggests the presence of underlying pathological processes or neurological damage. Although elevated brain age may not serve as a disease-specific marker but rather reflects a nonspecific indicator of cumulative pathological impacts on the brain, its strong association with disease risk provides critical insights for early identification of high-risk individuals. This research highlights the intricate interplay between systemic aging and neural circuitry, underscoring potential approaches for the prevention and early intervention of neuropsychiatric disorders.

Our study provides compelling evidence that the BAG independently predicts all-cause mortality within a large, population-based cohort. Three key findings highlight the clinical and prognostic relevance of BAG. First, each 1-year increase in BAG corresponded to a 12% elevated risk of mortality, even after adjusting for chronological age, sex, BMI, smoking status, and alcohol consumption. Second, a clear dose-response relationship was evident across BAG quartiles; participants in Q4 exhibited a 2.36-fold higher mortality risk compared to those in Q1. Third, nonlinear threshold effects were observed, indicating stable mortality risks when BAG was below 2 years, with significant increases beyond 3 years, reaching a HR greater than 2.0 between 6 to 8 years. Our findings position BAG as a valuable biomarker for biological aging with substantial prognostic implications. Our results align with emerging evidence linking accelerated biological aging to adverse health outcomes, including cardiovascular mortality and frailty progression. The 12% annual increase in mortality risk per year of elevated BAG mirrors previously reported associations between epigenetic age acceleration and cardiovascular mortality[87] (HR range: 1.58–1.59 per year). Additionally, the pronounced threshold effect at BAG values exceeding 6 years likely reflects cumulative neurological damage surpassing compensatory mechanisms. This interpretation is consistent with neuroimaging studies[88,89] reporting accelerated cortical thinning and increased white matter hyperintensities in individuals with higher BAG values. Overall, our findings underscore the importance of monitoring elevated BAG levels, offering potential targets for early interventions aimed at mitigating associated mortality risks.

This study further elucidates the strong association between healthy lifestyle factors and brain age. The results demonstrate that individuals who engage in regular physical activity, maintain a balanced diet, do not smoke, and consume alcohol in moderation tend to have a lower brain age compared to those with less healthy lifestyles. These findings suggest that healthy behaviors may help slow the brain aging process. Specifically, in the Q4 quartile of brain age distribution, non-smokers showed a reduction in BAG of 0.11 years compared to smokers ($p < 0.001$), while moderate alcohol consumption (BAG reduction of 0.20 years, $p < 0.001$) and regular exercise (BAG reduction of 0.14 years, $p = 0.026$) also demonstrated significant attenuation of neurodegenerative effects. These findings are highly consistent with recent research[90,91]. Among the different lifestyle subtypes, individuals in the metabolic risk-sedentary group showed the greatest brain age abnormalities, with statistically higher local brain age in the parahippocampal gyrus, thalamus, and middle frontal gyrus. These brain regions have been consistently associated with metabolic dysregulation and physical inactivity in previous studies, suggesting that this subtype may represent a high-risk state for accelerated brain aging. Furthermore, individuals in this group typically displayed higher BMI, consumption of high-fat and high-salt diets, physical inactivity, more severe depressive symptoms, and lower socioeconomic status, further supporting the potential mediating role of lifestyle factors in brain age changes. In contrast, the healthy lifestyle group, characterized by regular exercise, balanced nutrition, and lower levels of insomnia and loneliness, showed no statistically significant abnormalities in either global brain age or local brain age measures, and had the lowest depressive symptom scores. These results suggest that proactive lifestyle practices may exert protective effects against brain aging processes, potentially mitigating the acceleration of brain age.

The study primarily analyzed baseline BAG, with longitudinal trajectories examined only in subsets; future research should incorporate serial measurements to clarify dynamic associations with cognitive decline and neuropathology. Reliance on self-reported lifestyle data may introduce recall bias, and unmeasured confounders such as socioeconomic status or genetic risk could partially account for observed associations, underscoring the need for causal inference via Mendelian randomization or intervention trials. While BAG correlates with neuropsychiatric risk, its biological basis remains speculative and may benefit from integration with omics data to improve interpretability. The use of quartile-based stratification to identify high-risk subgroups (Q4) requires validation of actionable BAG thresholds in independent cohorts with clinical endpoints. Model performance may also be affected by demographic bias, as the 3D-ViT was trained predominantly on White British participants; applicability to underrepresented ethnicities and age extremes warrants further evaluation. Notwithstanding these limitations, the multi-cohort validation, nonlinear threshold analyses, and lifestyle-BAG interaction studies conducted herein enhance reproducibility and translational relevance, positioning BAG as a promising—though imperfect—biomarker in aging research.

Conclusion

This study systematically investigated the BAG as a biomarker of individual brain aging, defined as the difference between MRI-predicted brain age and chronological age. Elevated BAG values were robustly associated with heightened risk of neuropsychiatric disorders and all-cause mortality, highlighting its potential as a clinically informative tool for risk stratification.

Additionally, BAG demonstrated strong, domain-specific correlations with cognitive performance, particularly in memory, reasoning, and processing speed. Notably, BAG was also found to be modifiable: participants engaging in healthier lifestyle behaviors—such as regular physical activity and abstinence from smoking and alcohol consumption—exhibited attenuated brain aging trajectories. While a 3D Vision Transformer model facilitated accurate and generalizable BAG estimation across cohorts, the principal contribution of this work lies in establishing BAG as a practical, non-invasive indicator for assessing brain health, supporting early intervention, and guiding personalized strategies for promoting cognitive longevity.

Data availability

The source data underlying Figs. 1–5 are provided in Supplementary Data 1–7. Specifically, Fig. 1 data are available in Supplementary Data 1; Fig. 2 in Supplementary Data 2; Fig. 3b in Supplementary Data 3; Fig. 3c in Supplementary Data 4 and 5; Fig. 4 in Supplementary Data 6; and Fig. 5 in Supplementary Data 7. A data dictionary is also included in the Supplementary Data. The datasets used for model training and validation are available upon application from the UK Biobank (https://www.ukbiobank.ac.uk/enable-your-research/register), the Alzheimer's Disease Neuroimaging Initiative (ADNI; https://adni.loni.usc.edu/), and the Parkinson's Progression Markers Initiative (PPMI; https://www.ppmi-info.org/), subject to institutional approval and data use agreements.

Code availability

The source codes pertaining to both the brain age estimation model and data analysis in this manuscript are available at https://doi.org/10.5281/zenodo.16159489[92].

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Author contributions

Z.H. and K.W. jointly supervised the research. Z.H. and J.Z. conceptualized and designed the study. F.Y. developed the deep learning model. R.Z. and F.Y. performed model interpretation. R.Z., F.Y., and Z.H. conducted the data analysis and interpreted the results. R.Z., F.Y., H.M., Z.H., K.W., and J.Z. contributed to drafting the initial manuscript. All authors reviewed, revised, and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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